

File Handling

Answer the following questions to demonstrate your understanding of file handling concepts.

SECTION B: Short Answer Questions

Q1. Describe the steps involved in creating and writing to a new file in a programming language of your choice, including error handling for potential exceptions. (5 Marks)

Q2. Explain the difference between 'r+', 'w+', and 'a+' modes when opening a file for read and write operations, providing examples of when each mode is appropriately used. (5 Marks)

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

Short Questions - Grading Guide

Q1. Ideal Answer:

To create and write to a new file, first, import the necessary library (e.g., 'fs' in Node.js or 'io' in Python). Then, use a try-catch block to handle potential exceptions. Within the try block, specify the file path and name, and use a function like 'fs.writeFileSync()' or 'open()' to create the file. Next, write content to the file using a method like 'write()' or 'writeFileSync()'. In the catch block, handle exceptions such as 'Error' for general errors or 'TypeError' for type-related issues, logging or printing the error message.

Rubric:

Marks allocation: 1 mark for importing the library, 1 mark for using a try-catch block, 1 mark for specifying the file path and creating the file, 1 mark for writing content to the file, and 1 mark for handling exceptions in the catch block.

Q2. Ideal Answer:

The 'r+' mode allows for both reading and writing but starts at the beginning of the file. The 'w+' mode also allows for both reading and writing but truncates the file if it exists, starting at the beginning. The 'a+' mode allows for both reading and writing but starts at the end of the file if it exists, or creates a new file if it does not. 'r+' is used for updating existing content, 'w+' for overwriting or creating a new file, and 'a+' for appending to an existing file.

Rubric:

Marks allocation: 1 mark for explaining 'r+' mode, 1 mark for explaining 'w+' mode, 1 mark for explaining 'a+' mode, 1 mark for providing examples of appropriate use for each mode, and 1 mark for clarity and correctness of the explanation.